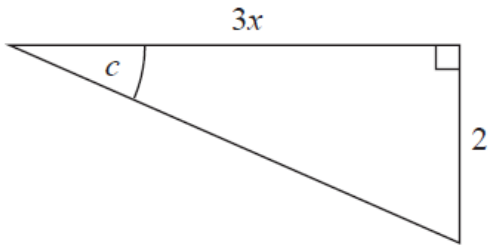
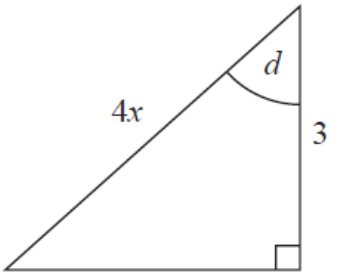
17 Here are two right-angled triangles.





The lengths of the sides are given in centimetres.

Given that

$$\tan c + \cos d = 1.5$$

find the size of angle c.
Give your answer correct to the nearest degree.

$\tan c = \frac{2}{3x}$ $\cos d = \frac{3}{4x}$

1 mark
Use O/A and A/H.

$\frac{2}{3x} + \frac{3}{4x} = 1.5$

Multiply by $12x$: $8 + 9 = 18x$

1 mark
Clear fractions.

$x = 17 \div 18$

$\tan c = 2 \div (3 \times x) = \frac{12}{17}$

1 mark
Use the found x.

$c = \tan^{-1}(12 \div 17) = 35.21759\ldots \approx 35°$

1 mark
35° nearest degree.

.....°

(Total for Question 17 is 4 marks)